

# Zhao (Dylan) Wang

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## EDUCATION

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### The Chinese University of Hong Kong (CUHK)

Ph.D. Student, Department of Computer Science and Engineering

Aug. 2021 – Jul. 2025 (expected)

Shatin, Hong Kong

- **Advisor:** Prof. [Qi Dou](#)

### Zhejiang University (ZJU)

B.Eng., College of Information Science and Electronic Engineering

Sep. 2017 – Jun. 2021

Hangzhou, China

- **Thesis:** Semi-supervised Medical Image Segmentation and Classification
- **Advisor:** Prof. [Honggang Zhang](#)

## RESEARCH INTERESTS

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My research interests lie in **computer vision** and **machine learning**, specifically, aiming to improve the Out-of-Distribution (OoD) generalization ability of deep models. Recently, I am investigating artificial intelligence generated content (AIGC), such as realistic videos and human poses.

## PUBLICATIONS

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### Google Scholar Profile

Note: \* indicates equal contributions.

#### Manuscript

- [M1] **Zhao Wang**, Chang Liu, Lingting Zhu, Shaoting Zhang, Qi Dou. "Improving Foundation Model for Endoscopy Video Analysis via Representation Learning on Long Sequences". *under review*, 2024. [\[paper\]](#)
- [M2] **Zhao Wang**, Aoxue Li, Enze Xie, Lingting Zhu, Yong Guo, Qi Dou, Zhenguo Li. "CustomVideo: Customizing Text-to-Video Generation with Multiple Subjects". *arXiv*, 2024. [\[paper\]](#)
- [M3] Qiao Deng, Zhongzhen Huang, Yunqi Wang, Zhichuan Wang, **Zhao Wang**, Xiaofan Zhang, Qi Dou, Yeung Yu Hui, Edward Hui. "Grounded Knowledge-Enhanced Medical VLP for Chest X-Ray". *under review*, 2024. [\[paper\]](#)
- [M4] Lingting Zhu, **Zhao Wang**, Zhenchao Jin, Guyin Lin, Lequan Yu. "Deformable Endoscopic Tissues Reconstruction with Gaussian Splatting". *arXiv*, 2024. [\[paper\]](#)
- [M5] Shizhan Gong, Yuan Zhong, Wenao Ma, Jinpeng Li, **Zhao Wang**, Jingyang Zhang, Pheng-Ann Heng, Qi Dou. "3DSAM-adapter: Holistic Adaptation of SAM from 2D to 3D for Promptable Medical Image Segmentation". *arXiv*, 2023. [\[paper\]](#)

#### Conference

- [C1] **Zhao Wang**, Aoxue Li, Zhenguo Li, Qi Dou. "Efficient Transferability Assessment for Selection of Pre-trained Detectors". *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2024. [\[paper\]](#)
- [C2] **Zhao Wang**, Aoxue Li, Fengwei Zhou, Zhenguo Li, Qi Dou. "Open-Vocabulary Object Detection with Meta Prompt Representation and Instance Contrastive Optimization". *British Machine Vision Conference (BMVC)*, 2023. [\[paper\]](#)
- [C3] **Zhao Wang\***, Chang Liu\*, Shaoting Zhang, Qi Dou. "Foundation Model for Endoscopy Video Analysis via Large-scale Self-supervised Pre-train". *International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, 2023. [\[paper\]](#) (early accept)
- [C4] Nan Lu, **Zhao Wang**, Xiaoxiao Li, Gang Niu, Qi Dou, Masashi Sugiyama. "Federated Learning from Only Unlabeled Data with Class-Conditional-Sharing Clients". *International Conference on Learning Representations (ICLR)*, 2022. [\[paper\]](#)

- [C5] Jiaping Hu, Chuanyang Zheng, Lijie Zhong, Keyan Yu, Yanjun Chen, **Zhao Wang**, Zhongping Zhang, Qi Dou, Xiaodong Zhang. "Predicting Knee Osteoarthritis Progression with a interpretable Deep Learning Approach on Magnetic Resonance Imaging: data from the Osteoarthritis Initiative". *International Society for Magnetic Resonance in Medicine Annual Meeting & Exhibition (ISMRM)*, 2022. [\[paper\]](#)
- [C6] Jiaping Hu\*, **Zhao Wang\***, Lijie Zhong, Keyan Yu, Yanjun Chen, Yingjie Mei, Qi Dou, Xiaodong Zhang. "Identification of Bone Marrow Lesions on Magnetic Resonance Imaging with Weakly Super-vised Deep Learning". *International Society for Magnetic Resonance in Medicine Annual Meeting & Exhibition (ISMRM)*, 2021. [\[paper\]](#)

### Journal

- [J1] Jiaping Hu, Chuanyang Zheng, Qingling Yu, Lijie Zhong, Keyan Yu, Yanjun Chen, **Zhao Wang**, Bin Zhang, Qi Dou, Xiaodong Zhang. "DeepKOA: A Deep-learning Model Predicts Progression in Knee Osteoarthritis Using Multimodal Magnetic Resonance Images from the Osteoarthritis Initiative". *Quantitative Imaging in Medicine and Surgery (QIMS, IF: 4.63)*, 2023. [\[paper\]](#)
- [J2] **Zhao Wang**, Quande Liu, Qi Dou. "Contrastive Cross-site Learning with Redesigned Net for COVID-19 CT Classification". *IEEE Journal of Biomedical and Health Informatics (JBHI, IF: 7.7)*, 2020. [\[paper\]](#)

### SELECTED AWARDS

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|---------------------------------------------------------------------------------------------------------|------|
| • Top 10 Research Nomination Award, ZJU                                                                 | 2021 |
| • <b>Highest</b> honor for only <b>top 20/50k+</b> students among all undergraduates and graduates      |      |
| • Honorable Winnner, Mathematical Contest in Modeling (MCM)                                             | 2020 |
| • 2 <sup>nd</sup> Prize, The 14th College Student Robot Competition, ZJU (Ranking: <b>6/200+</b> teams) | 2018 |
| • Northwest A & F University Scholarship (Ranking: <b>2/1000+</b> high school students)                 | 2016 |
| • 1 <sup>st</sup> Prize, Chinese Physics Olympiad (CPhO, Province Level, Top <b>0.1%</b> )              | 2016 |

### PROFESSIONAL SERVICES

#### Conference Reviewer:

- European Conference on Computer Vision (ECCV), 2024
- International Conference on Machine Learning (ICML), 2024
- International Conference on Learning Representations (ICLR), 2024
- IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), 2024
- Conference on Neural Information Processing Systems (NeurIPS), 2023
- British Machine Vision Conference (BMVC), 2023
- International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI), 2023, 2024
- International Symposium on Biomedical Imaging (ISBI), 2023, 2024
- International Conference on Information Processing in Computer-Assisted Interventions (IPCAI), 2022

#### Journal Reviewer:

- IEEE Transactions on Image Processing (TIP)
- IEEE Journal of Biomedical and Health Informatics (JBHI)
- Pattern Recognition Letters
- ACM Transactions on Computing for Healthcare
- Scientific Reports
- Frontiers in Artificial Intelligence
- Frontiers in Radiology
- IET Computer Vision

#### Program Committee:

- Distributed, Collaborative and Federated Learning Workshop@MICCAI, 2022, 2023

### Volunteer Helper:

- Medical Imaging meets NeurIPS, 2022

### Membership:

- IEEE Student Member
- ACM Student Member

## TEACHING

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### Teaching Assistant:

- CSCI1110 Problem Solving by Programming, CUHK Fall 2022
- CSCI1030 Hands-On Introduction to Java, CUHK Spring 2022
- CSCI3230 Fundamentals of Artificial Intelligence (with Elite-Mirror ESTR 3108), CUHK Fall 2021

## EXPERIENCE

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### Shanghai AI Laboratory

Research Intern, AI for Science Group

Nov. 2023 – Present  
Shanghai, China

- **Topic:** Whole-body Text-to-Pose Generation
- **Advisor:** Dr. [Shixiang Tang](#)

### SenseTime Group Limited

Research Intern, Smart City Group

Apr. 2021 – June 2021  
Hangzhou, China

- **Topic:** Domain Adaption, Representation Learning
- **Advisor:** Dr. [Rui Zhao](#), Dr. [Feng Zhu](#), Dr. [Shixiang Tang](#)

### University of California, San Diego (UCSD)

Visiting Student, Department of Electrical and Computer Engineering

Feb. 2020 – May. 2020  
La Jolla, CA

- **Topic:** Medical Imaging Generation
- **Advisor:** Prof. [Pengtao Xie](#)

## EXTRACURRICULAR ACTIVITIES

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### CC98 Forum, Zhejiang University

Technical Staff

Mar. 2019 – Jun. 2021  
Hangzhou, China

### Hangzhou East Railway Station

Volunteer

Winter, 2018  
Hangzhou, China

## SKILLS

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- **Programming Languages:** Python, C/C++, Bash, HTML/CSS
- **Tools and Frameworks:** PyTorch,  $\text{\LaTeX}$ , Git
- **Platforms:** Linux, MacOS, Windows